

Statistics for Linguistics

A practical introduction with R

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Preface

This coursebook serves as the companion text to a short introductory course on statistics for linguistics with R. The course is specifically aimed at linguists with little to no background in statistics or probability theory. Some basic knowledge of the R programming language is expected.

The following topics are covered:

- Descriptive statistics
- The (standard) normal distribution
- The sampling distribution
- Interval estimation
- Null hypothesis significance testing
 - t-tests
 - chi-squared test
 - linear regression

Covering both statistics and R in a workshop is a gargantuan endeavor and it's easy to feel lost along the way. Becoming a statistician in just three days? Don't even think about it. As if we linguists would be able to teach a language in barely two days - it can't and it won't happen. But that's ok. This course offers an introduction to the practice of statistics and this may hopefully give you a taste for more.

To reap the full benefits, you will have to do your “homework”, i.e., reread, recode, and redo the exercises. Or take another course or a more in depth tutorial on a specific topic.

Acknowledgements & License

This book was written in RStudio with [Quarto](#), “a multi-language, next generation version of R Markdown from Posit”. The book is licensed under the Creative Commons Attribution-NonCommercial-NoDerivs 4.0 License.

Good luck, and happy discoveries!

Ludovic De Cuyper

Chapter 1

Descriptive statistics

In this chapter, we will cover the foundational concepts of descriptive statistics, which help us summarize and explore datasets. Descriptive statistics provide insights into the structure and patterns of data before we move on to more complex analyses like hypothesis testing. We will cover both categorical and continuous variables, measures of central tendency, and measures of dispersion.

1.1 Variables

In a dataset, variables represent characteristics or properties that we measure. For example, a typical dataset in linguistics might look like this:

Participant	Group	L1	Score
1	A	monolingual	14
2	A	monolingual	15
3	A	bilingual	13
...	...	...	...
28	B	bilingual	15
29	B	multilingual	18
30	B	monolingual	9

Here, each row represents a participant, and the columns represent different variables, such as the participant's group, first language (L1), and their test score.

Types of variables

Variables are divided into two main types:

1. Categorical Variables:

- Also called qualitative variables, categorical variables represent distinct categories or groups.
- They have a limited number of values that are countable. For example, “Group” and “L1” are categorical variables.

2. Continuous Variables:

- Also called quantitative variables, continuous variables represent measurable quantities that can take any value within a range. For example, “Score” is a continuous variable because it can take any numerical value.

Your turn:

Identify whether the following are categorical or continuous variables:

- Sentence length (in number of words)
- Age (in years)
- Rating on a Likert scale

1.2 Univariate summary statistics

Univariate statistics summarize the distribution of a single variable. Depending on the type of variable, we summarize data differently.

Categorical variables

For categorical variables, we typically summarize data by calculating: - **Frequencies**: The number of times each category occurs. - **Proportions**: The relative frequency of each category as a percentage of the total.

Continuous Variables

For continuous variables, we summarize data using: - **Measures of Central Tendency**: Describes where the “center” of the data lies. These include the mean, median, and mode. - **Measures of Dispersion**: Describes how spread out the data is. These include the range, variance, and standard deviation.

1.3 Measures of central tendency

Sample mean

We consider a dataset with the heights of $N = 20$ basketball players.

```
height <- c(184, 217, 191, 196, 205, 177, 209, 185, 200, 189,
           221, 221, 194, 173, 196, 201, 188, 182, 189, 196)
height
```

```
[1] 184 217 191 196 205 177 209 185 200 189 221 221 194 173 196 201 188 182 189
[20] 196
```

The **mean** is the most commonly used measure of central tendency. It is the arithmetic average of all values in a dataset. The formula for the mean is:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n x_i$$

Where: - $\bar{X}$ is the mean, - n is the number of observations, and - x_i represents each individual value.

In R, you can compute the mean of a vector using:

```
mean(height)
```

```
[1] 195.7
```

Median

The median is a location value. A location value is defined as follows for a sample $X_{(1)}, \dots, X_{(n)}$:

$$\begin{aligned} X_{(1)} &= \min_{1 \leq i \leq n} X_i, \\ X_{(2)} &= \text{second smallest value } X_i, \\ &\vdots \\ X_{(n)} &= \max_{1 \leq i \leq n} X_i. \end{aligned}$$

We can sort a vector with:

```
sort(height, decreasing = FALSE)
```

```
[1] 173 177 182 184 185 188 189 189 191 194 196 196 196 200 201 205 209 217 221
[20] 221
```

The median is defined as in Equation 1.1:

$$M = \begin{cases} X_{((n+1)/2)} & \text{if } n \text{ uneven} \\ (X_{(n/2)} + X_{(n/2+1)})/2 & \text{if } n \text{ even} \end{cases} \quad (1.1)$$

- `height` includes $n = 20$ observations.
- The median is the average of the two middle values: $X_{(10)} = 194$ and $X_{(11)} = 196$. Hence: $(194 + 196)/2 = 195$.

Mode

The **mode** is the value that appears most frequently in a dataset. While it is less commonly used for continuous data, it is useful for identifying the most common category in categorical data.

To find the mode in R, you can use the `table()` function to find the frequency of each value.

1.4 Measures of dispersion

Measures of dispersion tell us how spread out the data is.

Location measurements

```
min(height)
```

```
[1] 173
```

```
median(height)
```

```
[1] 195
```

```
max(height)
```

```
[1] 221
```

```
range(height)
```

```
[1] 173 221
```

```
quantile(x = height, probs = c(0.25))
```

```
      25%  
187.25
```

```
quantile(x = height, probs = c(0.25, 0.75))
```

```
      25%      75%  
187.25 202.00
```

```
quantile(x = height, probs = c(0.10, 0.90))
```

```
      10%      90%  
181.5 217.4
```

Variance and standard deviation

The **variance** is a measure of the average squared deviation of each data point from the mean. It is calculated as:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{X})^2$$

You can think of the variance as the average (squared) distance between every observation and the mean.

The **standard deviation** is the square root of the variance:

$$s = \sqrt{s^2}$$

The standard deviation is useful because it has the same units as the original data, unlike the variance, which is in squared units.

In R, you can compute the variance and standard deviation with:

```
var(height) # Variance
```

```
[1] 185.3789
```

```
sd(height) # Standard deviation
```

```
[1] 13.61539
```

Mean Absolute Deviation (MAD)

The **mean absolute deviation** is another measure of dispersion, which calculates the average absolute deviation from the mean:

$$\text{MAD} = \frac{1}{n-1} \sum_{i=1}^n |x_i - \bar{X}|$$

1.5 Visualizing Data

Boxplot

A **boxplot** is a graphical representation of the distribution of a dataset. It shows the minimum, first quartile, median, third quartile, and maximum values.

You can create a boxplot in R with:

```
boxplot(height,  
  col = "#F1A42B",  
  ylab = "Height (cm)")
```

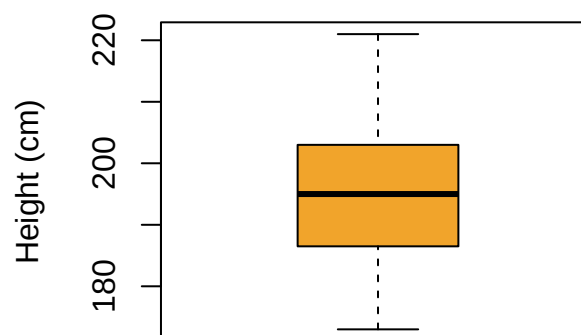


Figure 1.1: A boxplot

Histogram

A **histogram** is another way to visualize the distribution of continuous data. It shows the frequency of data points within specified intervals or “bins.”

You can create a histogram in R with:

```
hist(height,  
  col = "#F1A42B",  
  xlab = "Height (cm)")
```

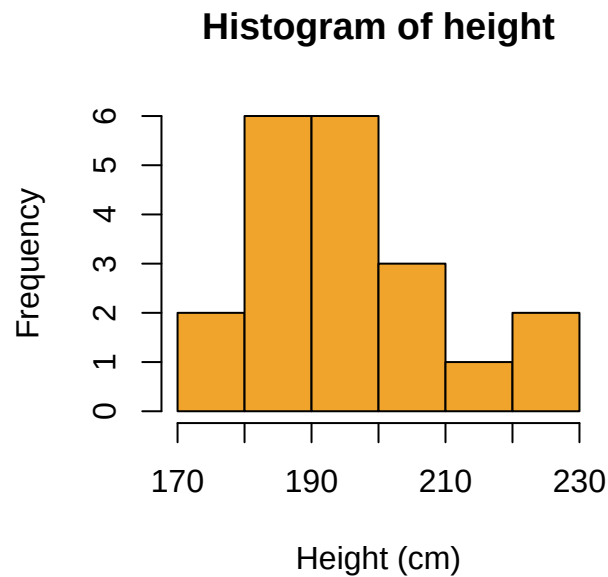


Figure 1.2: A histogram

You can also change the number of bins to get different views of the data, as in Figure 1.3:

```
hist(height,
      col = "#F1A42B",
      breaks = 6,
      main = "6 breaks",
      xlab = "Height (cm)")
hist(height,
      col = "#F1A42B",
      breaks = 40,
      main = "40 breaks",
      xlab = "Height (cm)")
```

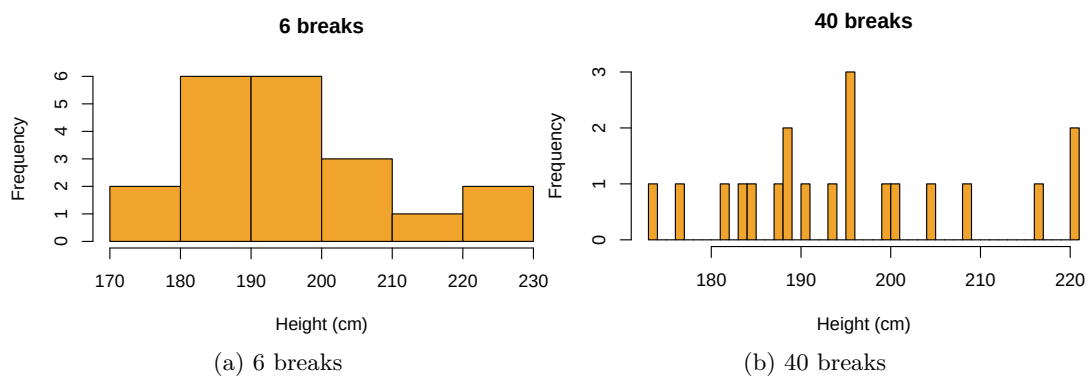


Figure 1.3: Same data, different bins

Skewed and bimodal distributions

Data can sometimes be skewed or have more than one mode. For example, the following code simulates a skewed distribution:

```
hist(exp(rnorm(100)), col = "#F1A42B", main = "", xlab = "")
```

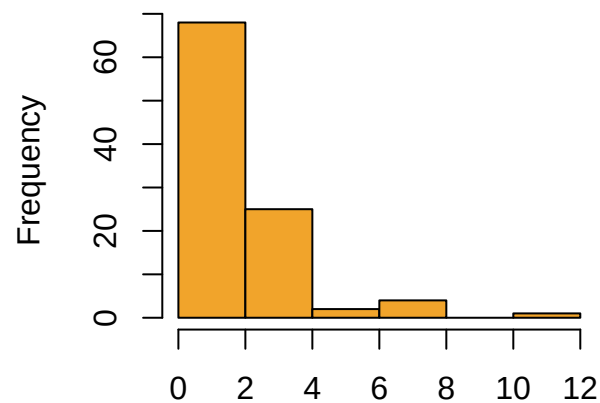


Figure 1.4: A skewed distribution

And the following code creates a bimodal distribution:

```
hist(c(rnorm(1000, mean = 30, sd = 2),  
      rnorm(1000, mean = 38, sd = 2)),  
     col = "#F1A42B", main = "", xlab = "")
```

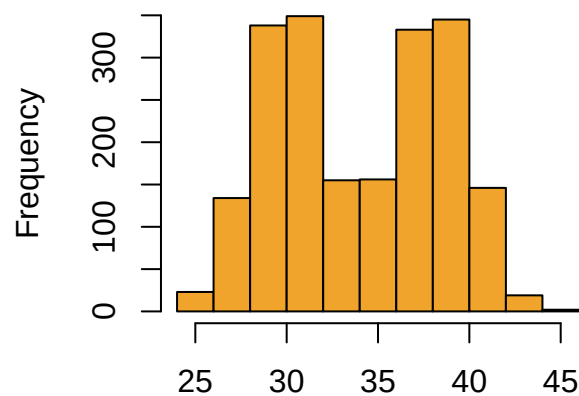


Figure 1.5: A bimodal distribution

Density curve

A **density curve** provides a smooth estimate of the distribution of a continuous variable. You can plot a density curve in R as follows:

```
ReactionTime <- rnorm(n = 1000, mean = 300, sd = 10)
densRT <- density(ReactionTime)
plot(densRT,
     main = "",
     ylab = "Density",
     xlab = "Reaction Time (ms)")
```

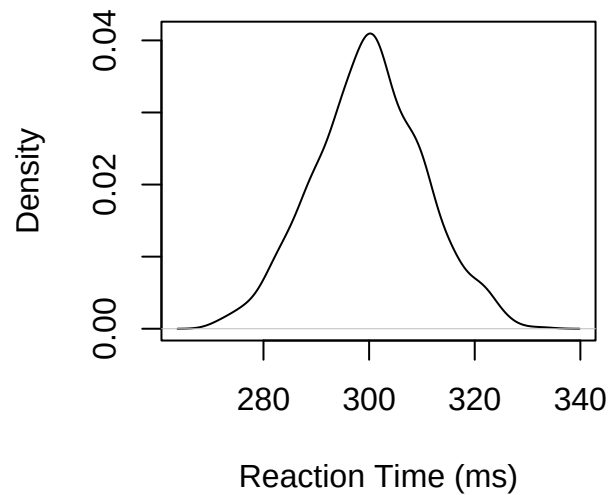


Figure 1.6: A density curve

Stripchart

A stripchart is handy when you have a smaller number of observations. Here we take a sample of 8 observations from the Reaction Time data.

```
sampleRT <- sample(x = ReactionTime, size = 8)
stripchart(sampleRT,
           xlab = "Reaction time (ms)",
           pch = 1)
```

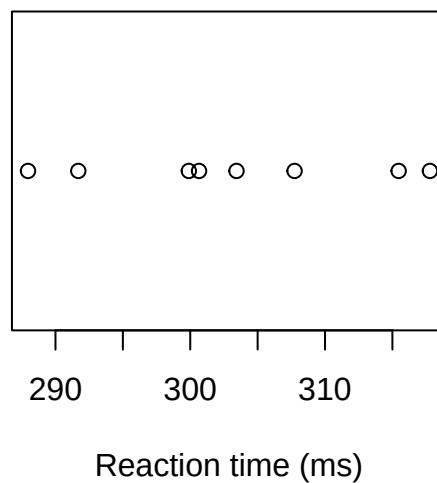


Figure 1.7: A stripchart

1.6 Categorical Data

To summarize categorical data, we often use frequency tables or proportions. For example, let's consider a dataset representing participants' educational level:

```
EduLevel <- c("high", "high", "ba", "high", "ba", "ma", "ma", "ma", "high", "ma")
```

First, we convert this character vector into a factor:

```
EduLevel <- as.factor(EduLevel)
str(EduLevel)
```

```
Factor w/ 3 levels "ba","high","ma": 2 2 1 2 1 3 3 3 2 3
```

`EduLevel` is now a factor variable with three categories or levels. R assigns a number to every category. Levels are assigned alphabetically. This can be overruled and the levels can be binned with the `levels()` function.

To summarize the data, use the `table()` function:

```
table(EduLevel)
```

```
EduLevel
  ba high  ma
   2   4   4
```

You can also calculate proportions using `prop.table()`:

```
prop.table(table(EduLevel))
```

```
EduLevel
  ba high  ma
0.2  0.4  0.4
```

Two continuous variables: scatterplot

Here is a mock dataset for ten participants who performed a pre- and posttest.

ID	Pretest	Posttest
1	13	15
2	8	9
3	11	16
4	12	13
5	16	17
6	15	17
7	11	12
8	6	10
9	10	13
10	14	15

A **scatterplot** is used to visualize the relationship between two continuous variables. For example, the relation between the pretest and posttest scores can be visualized as follows:

```
# recreate the data
Pretest <- c(13, 8, 11, 12, 16, 15, 11, 6, 10, 14)
Posttest <- c(15, 9, 16, 13, 17, 17, 12, 10, 13, 15)
plot(x = Pretest, y = Posttest, xlab = "Score Pretest", ylab = "Score Posttest")
```

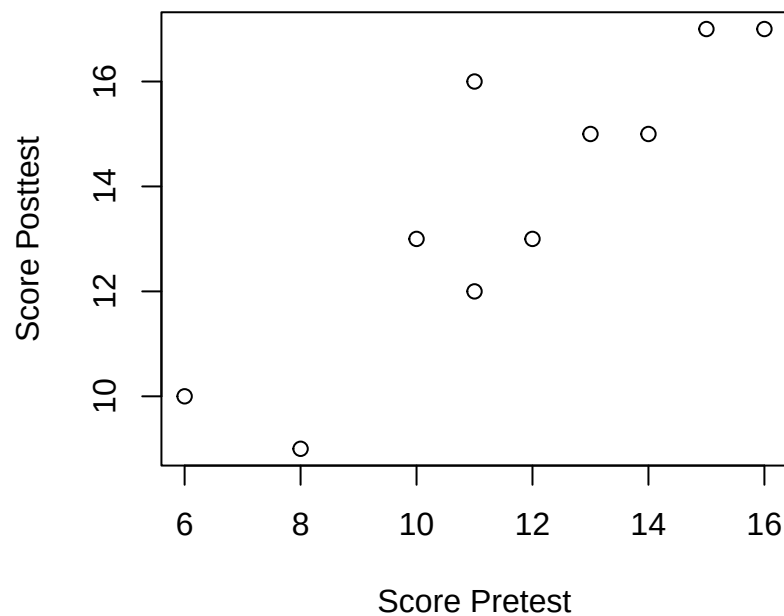


Figure 1.8: A scatterplot

You can also add a linear regression line to the scatterplot:

```
plot(x = Pretest, y = Posttest, xlab = "Score Pretest", ylab = "Score Posttest")
abline(lm(Posttest ~ Pretest))
```

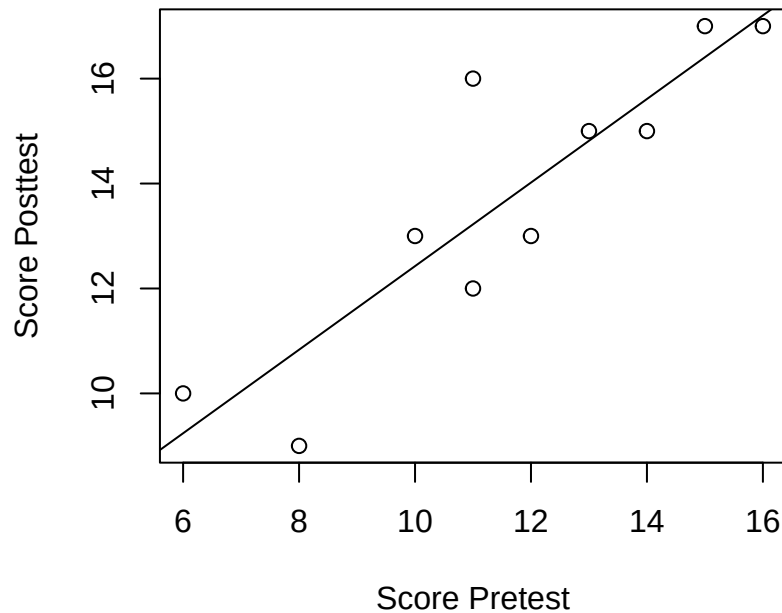


Figure 1.9: A scatterplot with regression line

Bivariate categorical data

Here is a mock dataset for ten speakers of English (native and non-native speakers) and whether they used a monophthong for the PRICE /a /vowel.

ID	Native	Monophthong
1	no	yes
2	no	yes
3	no	no
4	yes	no
5	yes	no
6	no	yes
7	no	no
8	no	no
9	yes	no
10	no	yes

We recreate the dataset as a dataframe:

```
Native <- c("no", "no", "no", "yes", "yes", "no", "no", "no", "yes", "no")
Mono <- c("yes", "yes", "yes", "yes", "no", "yes", "no", "no", "no", "yes")
cat <- data.frame(Native, Mono)
head(cat)
```

	Native	Mono
1	no	yes
2	no	yes
3	no	yes
4	yes	yes
5	yes	no
6	no	yes

We can summarize the bivariate relationship as a contingency table.

Bivariate table (1)

```
table("Native English" = cat$Native, # adding the variable gives a nicer table
      "Monophthong" = cat$Mono)
```

	Monophthong	
Native English	no	yes
no	2	5
yes	2	1

Bivariate table (2)

```
xtabs(~ Native + Mono, data = cat)
```

	Mono	
Native	no	yes
no	2	5
yes	2	1

Bivariate summary (3)

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2

-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
cat |> count(Native, Mono)
```

	Native	Mono	n
1	no	no	2
2	no	yes	5
3	yes	no	2
4	yes	yes	1

Bivariate summary (4)

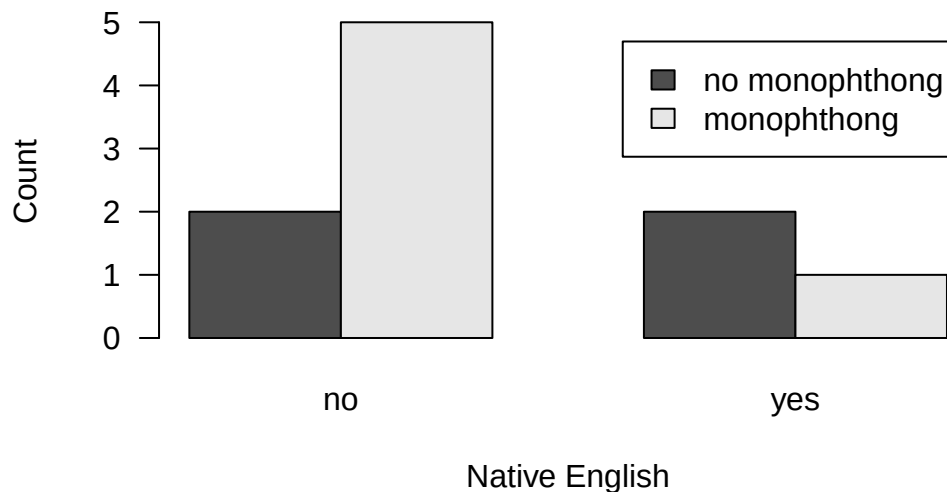
```
as.data.frame(table("Native English" = cat$Native,
                    "Monophthong" = cat$Mono))
```

	Native.English	Monophthong	Freq
1	no	no	2
2	yes	no	2
3	no	yes	5
4	yes	yes	1

Bivariate barplot

Bivariate tables are visualised by means of a clustered barplot.

```
barplot(table(cat$Mono, cat$Native),
        xlab = "Native English",
        ylab = "Count",
        legend.text = c("no monophthong", "monophthong"),
        beside = TRUE,
        las = 1)
```



1.7 Exercises

Exercise 1

We measured the average reaction time (in ms) of 20 Participants:

347	338	342	334
351	348	322	455
329	348	390	355
359	360	350	362
329	327	334	333

1. Create a vector in R called `RT` and assign the measurements to this object.
2. How many observations are there? (use a function, don't count)
3. What is the average Reaction Time?
4. Use R to calculate:
 - maximum
 - minimum
 - median
 - standard deviation
5. Visualise the distribution by means of:
 - a boxplot
 - a histogram

Exercise 2

Create the following data:

```
Pretest <- c(13, 8, 11, 12, 16, 15,
             11, 6, 10, 14)
Posttest <- c(15, 9, 16, 13, 17, 17,
              12, 10, 13, 15)
```

Ask Gemini to visualise the pre-posttest data by means of a spaghettiplot. Advise it to use tidyverse and ggplot2. You want to understand every line of code, so you also ask it to document every line of code.

1. What information is missing from this visualisation?

Exercise 3: dataset

For this exercise we use the phonetic data by Verbeke & Van Praet (2025)

ABSTRACT: This dataset contains the measurements of 210 checked steady-state vowels. Previous studies have shown that suprasegmental factors have an impact on the duration of vowels, such as prosodic boundary strength and the proximity to the end of an intonation unit boundary. To the previous findings, this study adds another potential factor, namely the difference between primary accented, secondary accented, and non-accented syllables. By diversifying between these factors, we seek to establish the actual influence that each factor might have on the duration steady-state vowels, which will be assessed using multiple linear mixed-effects regression models. R code for the data analysis is provided.

1. Import the dataset `Accentuation_Dataset.csv` in R.
2. Give a general summary of all the variables with the `summary()` function.
3. Summarize the variable `Duration_ms`:
 - mean
 - median
 - min
 - max
 - standard deviation
4. Visualise the variable `Duration_ms` by means of:
 - a histogram
 - a boxplot
5. Centering the data means to create a new variable by taking the difference between each observation and the mean of the variable. Center the data and create a histogram of this centered variable. Is the shape of the histogram different than the histogram of the uncentered data?
6. Summarize the variable `ACC` (Accentuation status). Give the frequencies and proportions of the levels.
7. Create a scatterplot to compare the measurements by the two annotators. Use `abline()` to add a regression line to visualize any linear trend.
8. Create clustered boxplots to compare `Duration_ms` by Accentuation status.
9. Use the function `xtabs()` to recreate table 1 from the README document shared in the dataset

Chapter 2

The normal distribution

In this chapter, we introduce the normal distribution, one of the most important probability distributions in statistics. The normal distribution is foundational for many statistical tests. We will explore what a normal distribution is, how to interpret it, and how to standardize data.

2.1 A “normal” distribution?

The normal distribution is a bell-shaped curve that is symmetric around the mean. It describes how values of a variable are distributed, with most observations clustering around the mean and fewer observations occurring as you move further from the mean.

The code below generates random samples from a normal distribution and visualizes them as histograms:

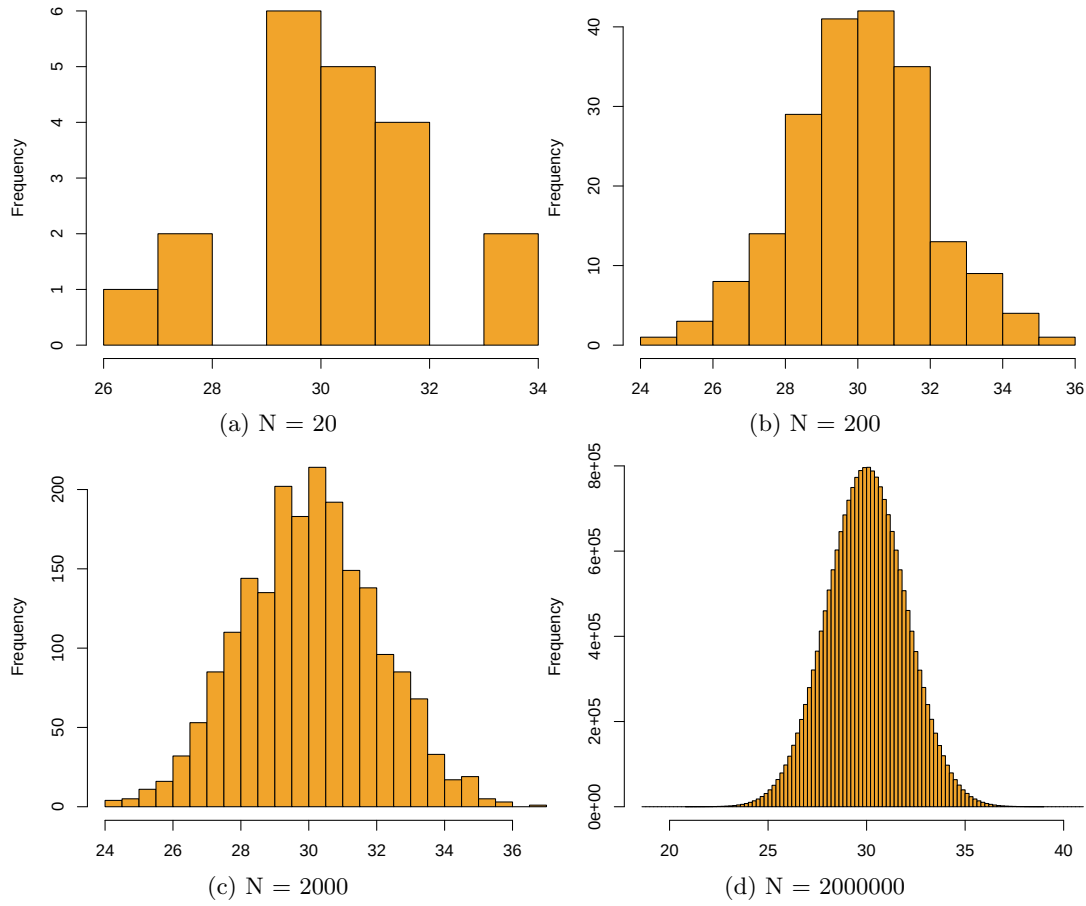


Figure 2.1: Normal distributions

As the sample size increases, the histogram looks more like the classic bell-shaped curve. This is a key feature of the normal distribution: larger samples better approximate the underlying distribution (if the sample is randomized!).

The normal density curve

The normal distribution is denoted as follows:

$$X \sim N(\mu, \sigma^2)$$

This reads: “X is a normally distributed variable with mean μ and variance σ^2 .”

The **density function** of the normal distribution is given by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

This formula may look complex, but it essentially tells us how the probability is distributed across the values of (X).

Visualizing the normal distribution

The following code visualizes a normal distribution with a mean of 180 and a standard deviation of 10, using R's `dnorm()` function:

```
plot(function(x)dnorm(x, mean = 180, sd = 10),  
      from = 140, to = 220,  
      col="#1E64C8", ylab = "", yaxt="n", xlab = "Height")
```

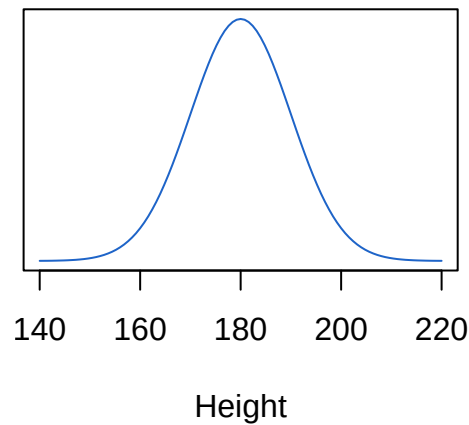


Figure 2.2: The normal density curve

Exploring changes in mean and variance

By changing the mean or the variance of the normal distribution, we can observe how the shape of the curve changes. Below, we plot three normal distributions with different means:

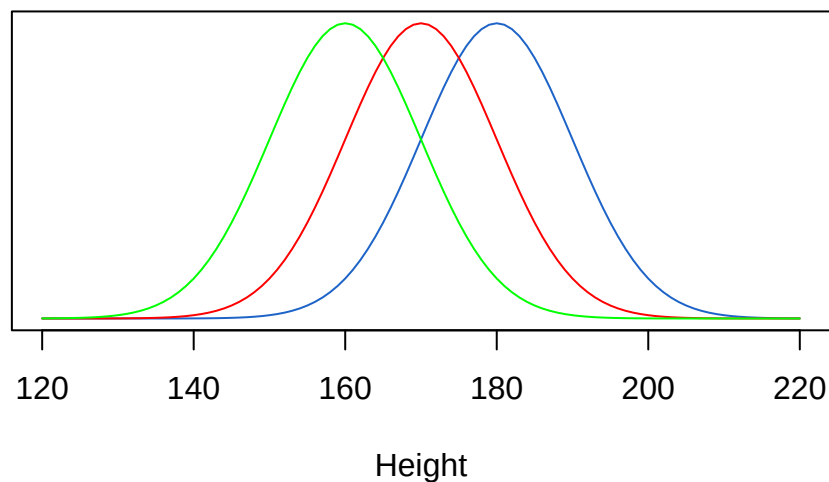


Figure 2.3: The normal distribution with difference means

Here, the variance remains the same, but the mean changes. The peak of the distribution shifts left or right based on the mean.

Similarly, we can visualize the effect of changing the variance while keeping the mean constant:

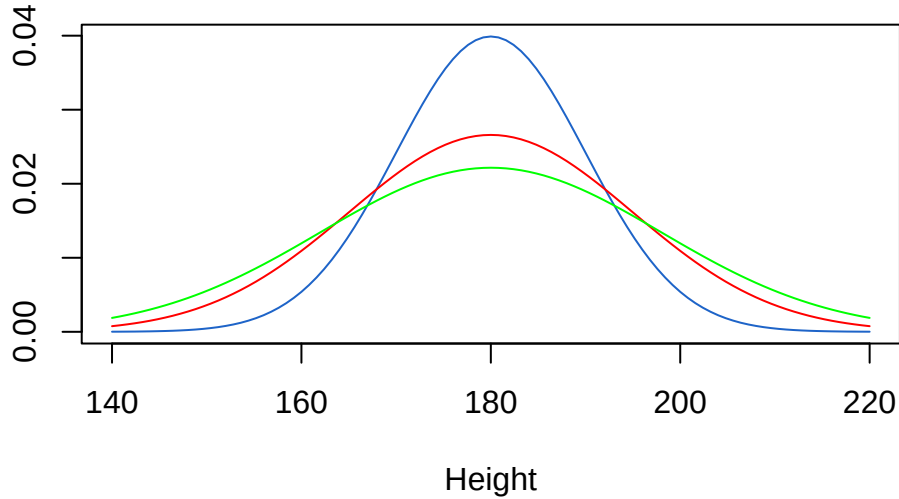


Figure 2.4: The normal distribution with different variances

As the variance increases, the distribution flattens, indicating that the data is more spread out.

Z-Score

Taking height' as an example, we intuitively know that someone who is 221 *cm* tall is very tall. We understand that this observation is exceptional or special. We can express 'exceptional' a bit more formally as: the probability that someone is even taller than 221 *cm* is very small. Formally, a probability is expressed as a percentage. So we can say that the percentage of people taller than 221 *cm* is very small. Let's assign a probability on purely hypothetical grounds (an educated guess): let's assume 1 in 1 million, or $\frac{1}{1,000,000}$ or 0.000001. Formally, we write: $P(\text{Height} > 221) = 0.000001$.

A **Z-score** tells us how many standard deviations an observation is from the mean. For example, someone who is 221 *cm* tall in a population where the mean height is 180 *cm* and the standard deviation is 10 *cm* has a very high Z-score. We can calculate this using:

```
z <- (221-180)/10
z
```

```
[1] 4.1
```

This Z-score tells us that a height of 221 *cm* is 4.1 standard deviations above the mean. In many cases, observations with Z-scores greater than 2 or 3 are considered unusual or exceptional.

Your turn

Calculate the z-score of the following values:

- 180

- 190
- 200
- 223
- 258

2.2 The standard normal distribution

Standardizing transforms a normal distribution into the **standard normal distribution**, which has a mean of 0 and a standard deviation of 1. This allows for comparisons across different datasets. The formulas for standardizing data are:

For a population:

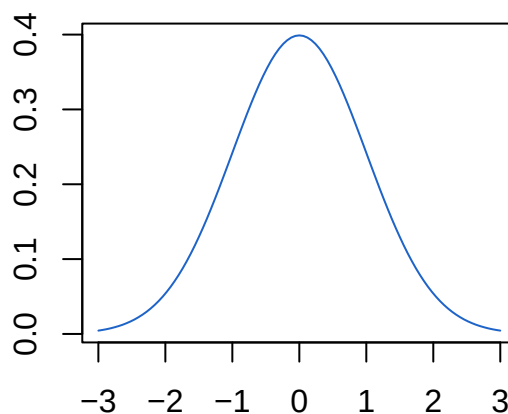
$$Z = \frac{X_i - \mu}{\sigma}$$

For a sample:

$$Z = \frac{X_i - \bar{X}}{s}$$

The **standard normal distribution** is a normal distribution with a mean of 0 and a standard deviation of 1. All normal distributions can be transformed into the standard normal distribution through standardization.

```
plot(function(x) dnorm(x, mean = 0, sd = 1),
      from = -3, to = 3, col="#1E64C8",
      ylab = "", xlab= "The standard normal distribution")
```



The standard normal distribution

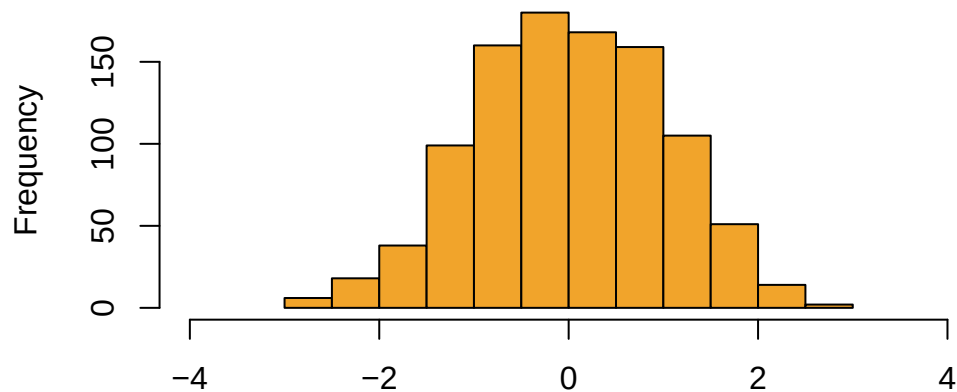
Figure 2.5: The standard normal distribution

Example

We draw a random sample of 1000 observations from a normal distribution with $\mu = 100$ and $\sigma = 13$. We standardise the sample and visualize by means of a histogram.

```
s <- rnorm(n = 1000, mean = 100, sd = 13) ①  
s_z <- scale(s, center = TRUE, scale = TRUE) ②  
hist(s_z, col = "#F1A42B", ③  
      main = "", ylab = "Frequency",  
      xlab = "",  
      xlim=c(-4,4))
```

- ① take a random sample;
- ② standardise the sample;
- ③ visualise the standardised sample with a histogram.



2.3 Cumulative proportions

In the context of the normal distribution, we can calculate the cumulative probability that a value falls below or above a certain threshold. For example, what is the probability that a person's height is greater than 160 cm in a population where the mean height is 180 cm and the standard deviation is 10 cm?

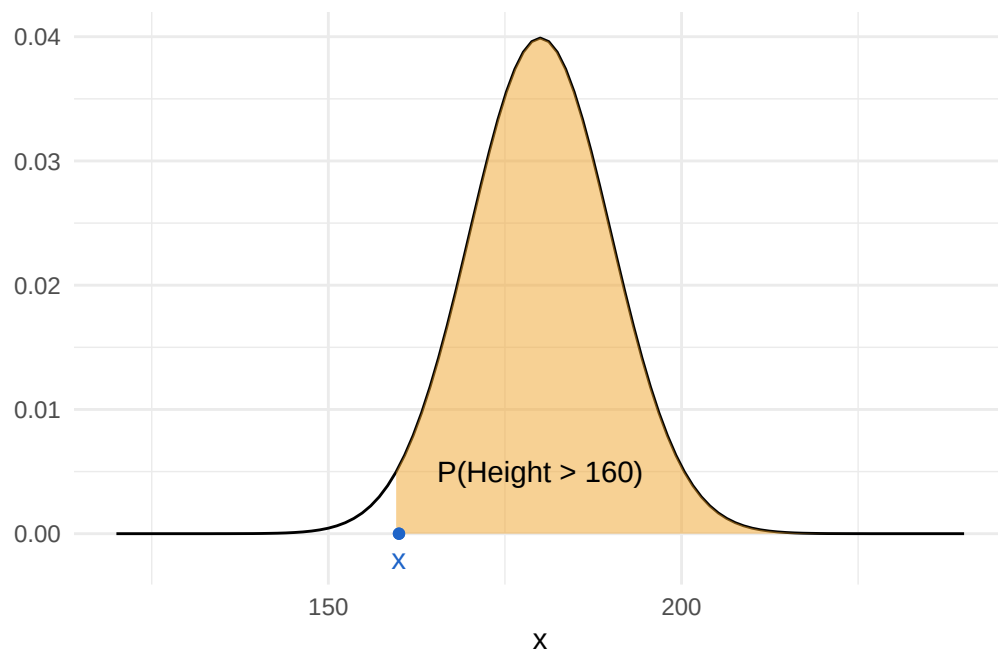


Figure 2.6: The normal density curve with a cumulative proportion

To calculate the proportion, we first calculate the Z-score, then we use the `pnorm()` function:

```
z <- (160-180)/10
pnorm(q = z, mean = 0, sd = 1, lower.tail = FALSE)
```

```
[1] 0.9772499
```

This shows the proportion of the population that has a height higher than 160 cm for the given normal distribution.

2.4 Finding critical values

Sometimes, we want to find the value that corresponds to a certain cumulative probability. Let's look at the distribution of height once again:

- Given: $Height \sim N(\mu = 180, \sigma = 10)$

What height corresponds to the 5th percentile (i.e., the smallest 5%)? Or:

- What is x for: $P(Height < x) = 5\%$?

We use the formula for the z-score:

$$z = \frac{x_i - \mu}{\sigma}$$

$$z * \sigma = x_i - \mu$$

$$z * \sigma + \mu = x_i$$

First we need to know z (the critical value on a standard normal distribution associated with the . We use the `qnorm()` function for this:

```
qnorm(p = 0.05,
      mean = 0,
      sd = 1,
      lower.tail = TRUE)
```

```
[1] -1.644854
```

Now we plug everything we know into the formula $x_i = \mu + z * \sigma$:

$$\begin{aligned} x_i &= z * \sigma + \mu \\ &= -1.64 * 10 + 180 \\ &= 163.3 \end{aligned}$$

5% of all values on a normal distribution with $\mu = 180$ and $\sigma = 10$ is smaller than 163.6 *cm*.

2.5 Checking for normality

In practice, we often need to check whether a dataset is normally distributed before applying certain statistical tests. Common ways to check for normality include:

1. Thinking about the data generating proces.
2. Visual inspection (e.g., histograms, QQ-plots).
3. Statistical tests (e.g., Shapiro-Wilk test).

QQ-Plot

```
Height<-c(184,217,191,196,205,177,209,185,200,189,
          221,221,194,173,196,201,188,182,189,196)
```

Let's look at a histogram first:

```
library(ggplot2)
df <- data.frame(Height)
ggplot(df, aes(x = Height)) +
  geom_histogram(bins = 5,
                 fill = "#F1A42B",
                 aes(y = after_stat(density))) +
  geom_density() +
  theme_minimal()
```

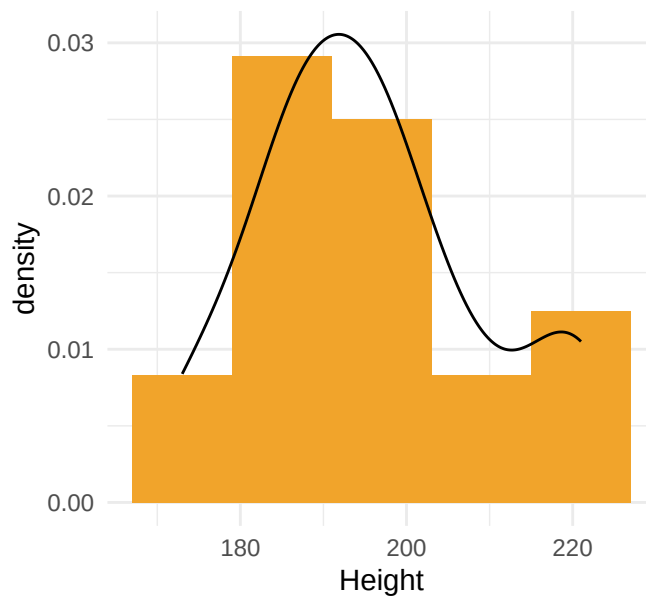


Figure 2.7: A histogram of Height

A **QQ-plot** compares the quantiles of the data with the quantiles of a normal distribution. If the data is normally distributed, the points should lie along the diagonal line.

```
ggplot(df,
  aes(sample = Height)) +
  stat_qq() +
  stat_qq_line() +
  theme_minimal()
```

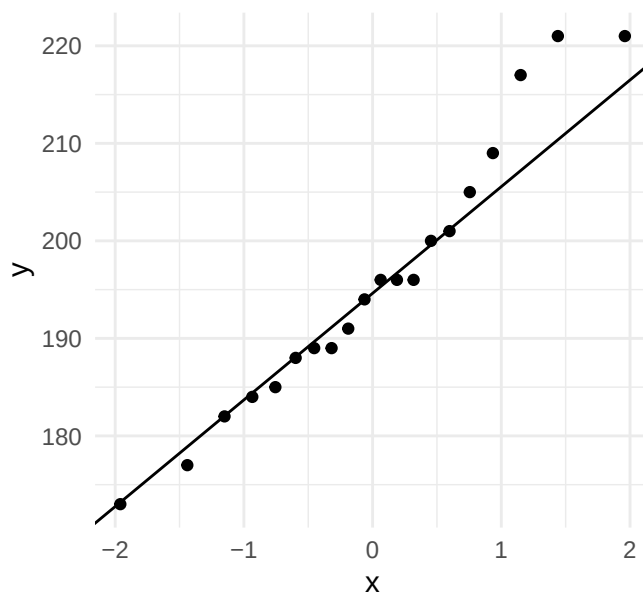


Figure 2.8: A qq-plot

Shapiro-Wilk Test

The **Shapiro-Wilk test** is a formal test for normality. A significant result (p-value < 0.05) suggests that the data is not normally distributed.

```
shapiro.test(x = Height)
```

```
Shapiro-Wilk normality test
```

```
data: Height
```

```
W = 0.95503, p-value = 0.4499
```

2.6 Exercises

Exercise 4

Let's look at the heights of the 20 basketball players again.

```
height <- c(184, 217, 191, 196, 205, 177, 209, 185, 200, 189,
            221, 221, 194, 173, 196, 201, 188, 182, 189, 196)
```

- Standardise the Height variable. Call this new variable Height_z.
- Interpret the new variable.
- What is the average of Height_z?

Exercise 5

IQ-measurements are normally distributed with $\mu = 100$ and $\sigma = 15$. What percentage of the population has:

- $IQ < 90$
- $70 < IQ < 130$
- $105 < IQ < 120$
- $P(IQ > 100)$?

Exercise 6

1. Is Duration_ms from Verbeke & Van Praet (2025) normally distributed?

Chapter 3

The sampling distribution

We start by asking a critical question for understanding statistics:

What would happen if we took multiple samples?

Imagine we are interested in estimating the mean height of a population where height is normally distributed with a mean of $\mu = 180$ and a standard deviation of $\sigma = 10$.

The following example shows how to take three random samples from this population and compute the mean for each sample.

```
S_1 <- rnorm(n = 200, mean = 180, sd = 10)
samplemean_1 <- mean(S_1)
samplemean_1
```

```
[1] 180.1239
```

```
S_2 <- rnorm(n = 200, mean = 180, sd = 10)
samplemean_2 <- mean(S_2)
samplemean_2
```

```
[1] 179.7169
```

```
S_3 <- rnorm(n = 200, mean = 180, sd = 10)
samplemean_3 <- mean(S_3)
samplemean_3
```

```
[1] 180.3462
```

These sample means are estimates of the population mean, and by repeating this process, we can observe how the sample means vary.

3.1 The sampling distribution

Now, let's extend this by taking **2000 samples**.

The sampling distribution of the mean refers to the distribution of the means of all possible samples of a fixed size from the population.

```
S_2000 <- replicate(n = 2000,
                    expr = rnorm(n = 200,
                                mean = 180,
                                sd = 10))
S_2000[1:10, 1:5]
```

```
      [,1]      [,2]      [,3]      [,4]      [,5]
[1,] 184.8727 174.9412 174.3829 190.0285 166.9872
[2,] 189.6921 183.4087 175.8190 164.9498 165.7813
[3,] 177.5399 188.3565 189.1355 170.9369 155.8821
[4,] 187.4622 177.8679 185.0698 195.6500 172.0940
[5,] 188.4646 169.3528 172.1943 179.2306 170.0823
[6,] 174.6542 176.9524 167.9431 177.1528 171.7258
[7,] 160.2988 160.1652 198.9418 170.5884 184.8843
[8,] 181.8225 188.2669 179.8145 175.7587 203.9556
[9,] 185.8224 172.0178 199.4861 194.0366 172.5177
[10,] 172.5833 184.6098 180.9487 160.8794 171.0813
```

In this example, `S_2000` is a matrix where each column represents a sample. The next step is to calculate the mean for each sample and display the first 20 sample means.

```
S_means <- colMeans(S_2000)
head(S_means, 20)
```

```
[1] 180.7482 179.5381 179.8986 179.4382 179.4374 178.8400 179.7541 179.2652
[9] 179.6312 179.3901 179.0465 180.1085 179.8836 180.2251 178.7342 179.7838
[17] 179.3822 179.9531 179.9109 179.4487
```

3.2 Visualizing the sampling distribution

We can now visualize the sampling distribution of the mean using a histogram:

```
hist(S_means, col = "#1E64C8",
     main = "",
     ylab = "Frequency",
     xlab = "sample mean",
     las = 1,
     breaks = 20)
```

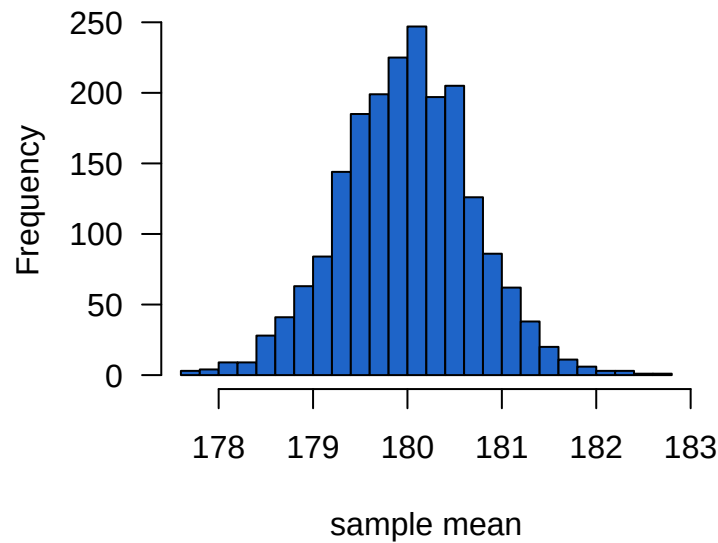


Figure 3.1: The sampling distribution of the mean

The sampling distribution of the mean typically follows a normal distribution, regardless of the population distribution, as long as the sample size is large enough (according to the **Central Limit Theorem**).

3.3 The sampling vs. the population distribution

We can now compare the sampling distribution of the mean to the population distribution.

```
plot(function(x) dnorm(x, mean = 180, sd = 10),
      from = 150, to = 210,
      col = "#1E64C8",
      lwd = 1.4,
      xlab = "Height",
      ylab = "density",
      ylim = c(0,0.6),
      las = 1)
hist(S_means, col = "#F1A42B",
      main = "",
      ylab = "Frequency",
      xlab = "Sample mean",
      xlim = c(150, 210),
      las = 1,
      breaks = 12, freq = FALSE, add = TRUE)
```

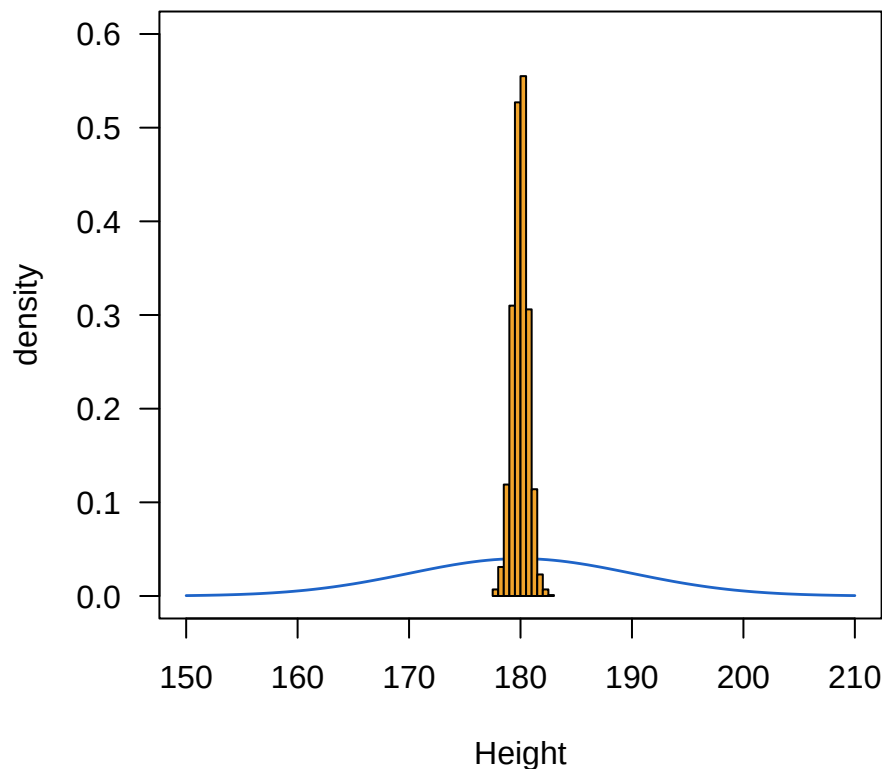


Figure 3.2: The sampling and population distribution

3.4 Standard Error

The **standard deviation** of the population is $\sigma = 10$, but the standard deviation of the sample means (called the **standard error**) is smaller:

```
sd(S_means)
```

```
[1] 0.6988478
```

The standard error SE is calculated as:

$$SE = \frac{\sigma}{\sqrt{n}}$$

where n is the sample size. If σ is unknown, we estimate it using the sample standard deviation s :

$$SE = \frac{s}{\sqrt{n}}$$

3.5 The t-Distribution

In most cases, we will not know σ . If we estimate σ by s (the sample standard deviation), the standardized sampling distribution of the mean no longer follows a normal distribution, but a **t-distribution** with $(n-1)$ degrees of freedom (abbreviated as ‘df’):

$$\frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \sim t_{n-1}$$

We can visualize the t-distribution for different degrees of freedom (df) and compare it with the standard normal distribution:

```
plot(function(x) dnorm(x), -3, 3, col = "black", ylab = "")
plot(function(x) dt(x, df = 1), -3, 3, col = "red", add = TRUE, ylab = "")
plot(function(x) dt(x, df = 10), -3, 3, col = "#1E64C8", add = TRUE, ylab = "")
plot(function(x) dt(x, df = 30), -3, 3, col = "green", add = TRUE, ylab = "")
legend(-3.2, 0.41, legend = c("df = 1", "df = 10", "df = 30", "N(0,1)"),
      col = c("red", "#1E64C8", "green", "black"), lty = 1, cex = 0.6, box.lty = 0)
```

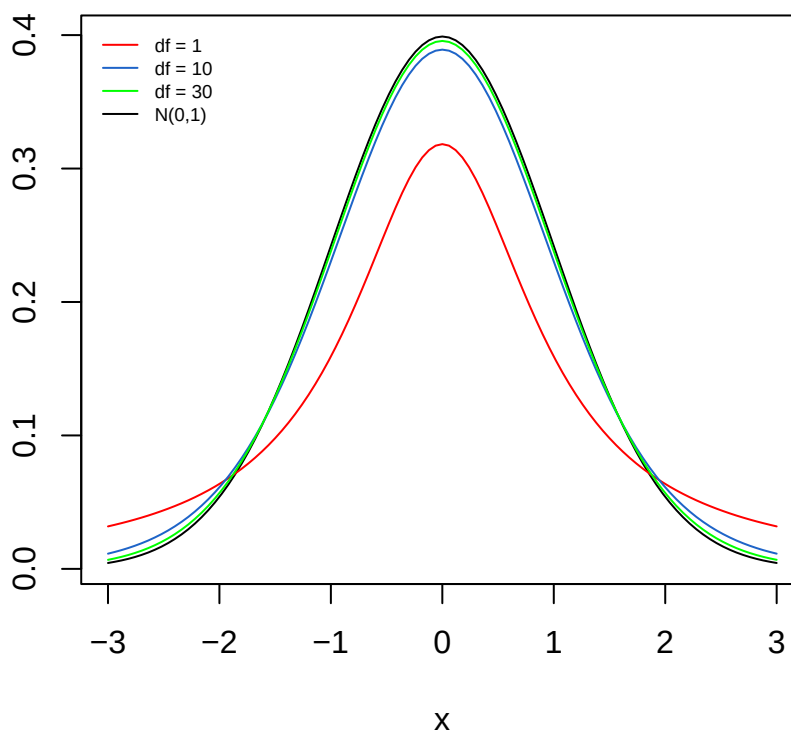


Figure 3.3: The t-distribution

3.6 R Functions for the t-Distribution

In R, we can sample from a t-distribution using `rt()`:

```
s <- rt(n = 1000, df = 59)
head(s)
```

```
[1] -1.0159335  0.4434715  0.2883927 -0.2194852  0.2397600  0.6238966
```

We can also compute the cumulative probability for a given t-value using `pt()`:

```
pt(q = -1.2, df = 60, lower.tail = TRUE)
```

```
[1] 0.1174281
```

To find the critical values for a given significance level, we use `qt()`:

```
qt(p = c(0.025, 0.975), df = 40)
```

```
[1] -2.021075  2.021075
```

3.7 Confidence interval for the mean

So far, we have calculated point estimators:

(Population)parameter	(Sample)statistic
μ	$\bar{X}$
σ	s

We can also calculate an interval (two numbers) rather than one point (one number). Those two numbers are then called a confidence interval. A confidence interval is estimated based on the sampling distribution of the mean.

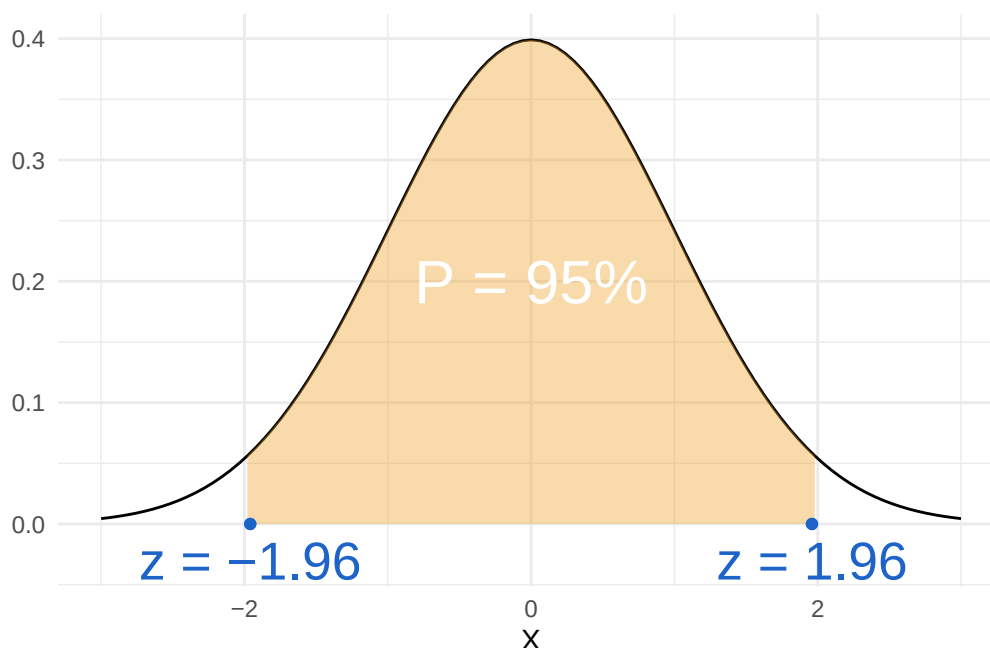


Figure 3.4: The sampling distribution of the mean

A confidence interval provides a range of values that is likely to contain the population parameter. The formula for a confidence interval is:

$$\text{lower bound} = \bar{X} - z_{\alpha/2} \cdot SE$$

$$\text{upper bound} = \bar{X} + z_{\alpha/2} \cdot SE$$

For example, if $N = 40$, $\sigma = 3$, and $\bar{X} = 41$, the confidence interval can be computed as:

```
X_bar <- 41
n <- 40
sigma <- 3
SE <- sigma / sqrt(n)
Z_alpha <- abs(qnorm(p = 0.5 / 2))
lower <- X_bar - Z_alpha * SE
upper <- X_bar + Z_alpha * SE
c(lower, upper)
```

```
[1] 40.68006 41.31994
```

3.8 Confidence interval with unknown variance

In most circumstances, the variance is unknown. In that case, the parameter needs to be estimated from the sample; s^2 estimates σ^2 .

For example, we draw a sample from a normal distribution and pretend we don't know the population variance.

```
set.seed(4635)
dd <- rnorm(n = 40, mean = 41, sd = 3) # we take a sample from a population with sigma = 3
head(dd, 20)
```

```
[1] 37.68313 42.80806 41.31365 46.18608 39.60299 40.31846 40.82984 39.37190
[9] 38.87229 41.44719 36.89051 41.90103 47.36603 41.36113 40.41350 38.00208
[17] 46.64815 36.89039 46.52108 42.28382
```

The variance from the sample (s^2) is equal to:

```
s2 <- var(dd)
```

This sample standard deviation s^2 is an estimator for the population standard deviation σ^2 .

As a “penalty” for estimating the variance (or standard deviation), we use the t-distribution to calculate the confidence interval of the mean:

```
X_bar <- mean(dd)
n <- 40
s <- sd(dd)
SE <- s / sqrt(n)
t_alpha <- abs(qt(p = 0.025, df = n - 1)) # note: qt() rather than qnorm()
lower <- X_bar - t_alpha * SE
upper <- X_bar + t_alpha * SE
c(lower, upper)
```

```
[1] 40.20862 41.89450
```

3.9 Exercises

Exercise 7

What is the 95% CI for `Duration_ms` from Verbeke & Van Praet (2025)?

Chapter 4

Null Hypothesis Significance Testing (NHST)

In this chapter, we introduce the concept of **null hypothesis significance testing (NHST)**, focusing on its practical application. The theory behind NHST is kept to a minimum, as the goal is to familiarize you with interpreting R output for various statistical tests. The next chapter will delve deeper into the theory behind NHST.

4.1 The procedure

The procedure for conducting a hypothesis test consists of the following steps:

1. **Formulate the hypotheses:** Define the null hypothesis (H_0) and the alternative hypothesis (H_A).
2. **Choose a significance level:** Typically, ($\alpha = 0.05$) is used, but other levels can be chosen.
3. **Choose the appropriate test:** Select the test based on your data and research question.
4. **Perform the test:** Ensure that the assumptions of the test are met.
5. **Interpret the findings:** Look at the p-value and decide whether to reject the null hypothesis.

The goal of hypothesis testing is to compare a difference or measure an effect, while **taking into account** variability. This variability reflects what would happen if we repeated the experiment or sampling process multiple times. In statistical terms, variability is captured by the **Standard Error**, which quantifies how much sample estimates (like the mean) fluctuate from one sample to another.

The outcome of any statistical test, in the frequentist paradigm, is a **test statistic**, which measures the difference or effect relative to the variability in the data. A key question in hypothesis testing is: **How “special” is my result?** In other words, we want to know how extreme or unusual the observed test statistic is, under the assumption that the null hypothesis is true.

The answer to this question lies in the **P-value** of the test statistic. The P-value tells us the probability of obtaining a result at least as extreme as the one observed, given that the null hypothesis is true. In frequentist statistics, the test statistic is often analogous to a **z-score**, which tells us how many standard deviations an observation is from the mean. The

P-value can be thought of as the cumulative proportion of the distribution that lies beyond the observed test statistic.

4.2 Examples of basic statistical tests

Here are some common statistical tests you will encounter in practice, each of which can be performed using R:

- **Two-sample t-test:** Test the difference between two independent means.
- **One-sample t-test:** Compare a sample mean to a known value.
- **Paired t-test:** Compare two related means.
- **Chi-squared test:** Test the association between two categorical variables.
- **Linear regression:** Examine the linear relationship between two continuous variables.

Each test is associated with assumptions and produces key outputs like the test statistic, degrees of freedom, and p-value.

4.3 Two-sample t-test

Example

We draw two samples from two different populations. We assume that both populations are equally and normally distributed. We simulate a difference in means.

```
Group <- c(rep("A", 1000), rep("B", 1000))
Height <- c(rnorm(1000, 180, 10), rnorm(1000, 160, 10))
MyData <- data.frame(Group, Height)
head(MyData)
```

	Group	Height
1	A	190.2297
2	A	177.7275
3	A	181.4999
4	A	179.1730
5	A	176.0523
6	A	185.0038

Then we visualize the data with overlapping histograms:

```
library(ggplot2)
ggplot(MyData, aes(x=Height, fill=Group)) +
  geom_histogram(position="identity", alpha=0.4)
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

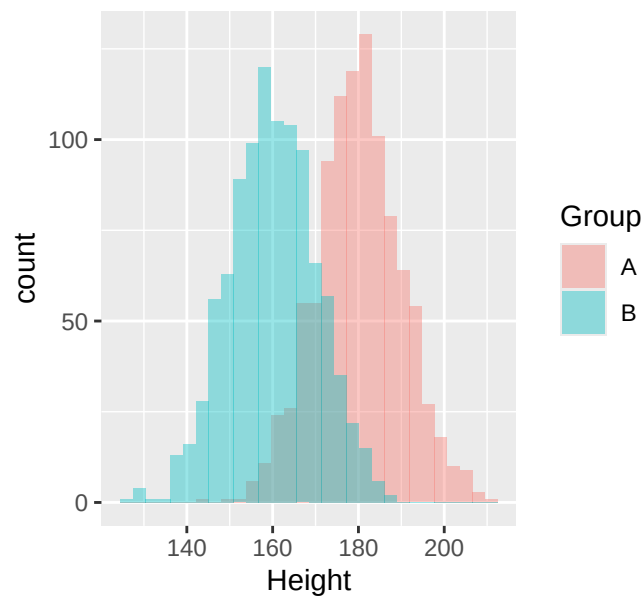


Figure 4.1: Height by Group by means of overlapping histograms

Performing the two-sample t-Test

To test if there is a significant difference between the means of the two groups, we use the `t.test()` function:

```
t.test(MyData$Height ~ MyData$Group, var.equal=TRUE)
```

Two Sample t-test

data: MyData\$Height by MyData\$Group

t = 44.615, df = 1998, p-value < 2.2e-16

alternative hypothesis: true difference in means between group A and group B is not equal to 0

95 percent confidence interval:

19.17290 20.93596

sample estimates:

mean in group A mean in group B

180.0697 160.0152

The P-value is smaller than 5%, so the test is statistically significant.

4.4 One-sample t-test

Example

In a one-sample t-test, we test whether the mean of a sample is different from a known value (e.g., is the average height different from 166 cm?):

```
mean(MyData$Height)
```

```
[1] 170.0425
```

Now, let's perform the one-sample t-test:

```
t.test(MyData$Height, mu = 166)
```

One Sample t-test

```
data: MyData$Height
t = 12.734, df = 1999, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 166
95 percent confidence interval:
 169.4199 170.6651
sample estimates:
mean of x
 170.0425
```

Again, the P-value is smaller than 5%, so we decide to reject the null hypothesis. Based on a one-sample t-test we found a statistically significant difference between the average Height of 169.7 and the expected average of 166 ($t = 11.9$, $df = 1999$, $P\text{-value} < 0.001$).

4.5 How to report a statistical result?

There are different ways to report the results of a statistical test/model. In an article you should definitely not write things like “we decide to reject the null hypothesis...”. This is statistical parlance that is only used in a statistics class or a report.

The APA guidelines can be found at owl.purdue.edu.

The best advice is to follow the guidelines of the specific journal in which you would like to publish your article and to emulate the style and register of highly ranked articles.

4.6 Chi-squared test

Next, we look at a chi-squared test, which tests for an association between two categorical variables. In this example, we examine the relationship between dyslexia status and font preference in a dataset:

```
dyslexia <- read.csv2("dyslex.csv")
tbl <- table(dyslexia$Dyslexi, dyslexia$Preference)
tbl
```

	Arial	Dyslexie	TimesNewRoman
no	53	48	14
yes	29	26	2

We can visualize this relationship with a clustered bar plot:

```
barplot(tbl, beside = TRUE, col=c("green", "red"), las=1)
legend(6, 50, legend=c("dyslexic", "not dyslexic"),
      box.lty=0, cex=1.5, fill=c("red", "green"))
```

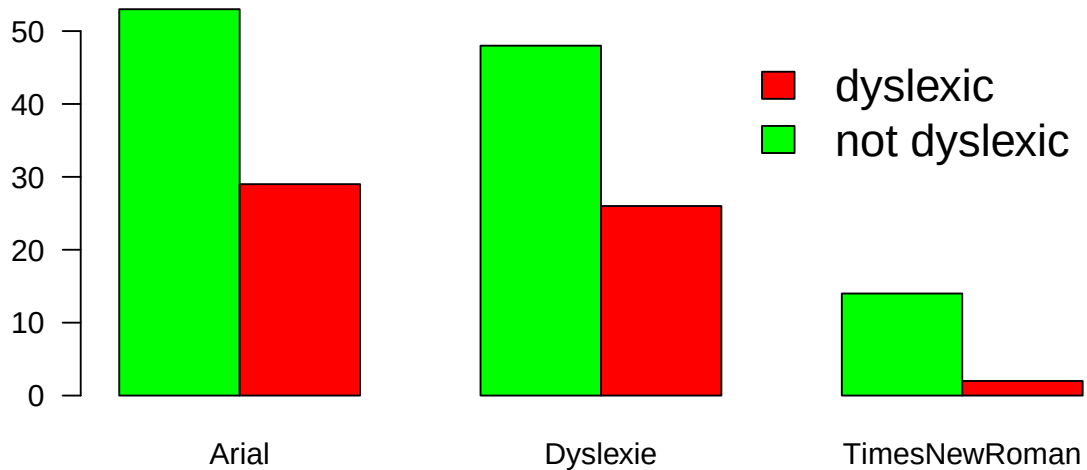


Figure 4.2: A clustered barplot of Font Preference by Dyslexia status

Performing the chi-squared test

To evaluate the association between dyslexia and font preference, we use the `chisq.test()` function:

```
chisq.test(tbl)
```

Pearson's Chi-squared test

```
data: tbl
X-squared = 3.3926, df = 2, p-value = 0.1834
```

If the p-value is above 0.05, we fail to reject the null hypothesis, meaning there is no significant association between dyslexia status and font preference.

4.7 Simple Linear Regression

We now examine the relationship between two continuous variables using **simple linear regression**. The `gcookbook::heightweight` dataset contains different body measurements. First, let's inspect the structure of the dataset:

```
library(gcookbook)
str(heightweight)
```

```
'data.frame': 236 obs. of 5 variables:
 $ sex      : Factor w/ 2 levels "f","m": 1 1 1 1 1 1 1 1 1 1 ...
```

```
$ ageYear : num  11.9 12.9 12.8 13.4 15.9 ...
$ ageMonth: int  143 155 153 161 191 171 185 142 160 140 ...
$ heightIn: num  56.3 62.3 63.3 59 62.5 62.5 59 56.5 62 53.8 ...
$ weightLb: num  85 105 108 92 112 ...
```

Visualizing the relationship

To examine the relationship between age and height, we start by plotting a scatterplot:

```
p <- ggplot(heightweight, aes(x=ageYear, y=heightIn)) +
  geom_point(col="steelblue", size=2.5)
p
```

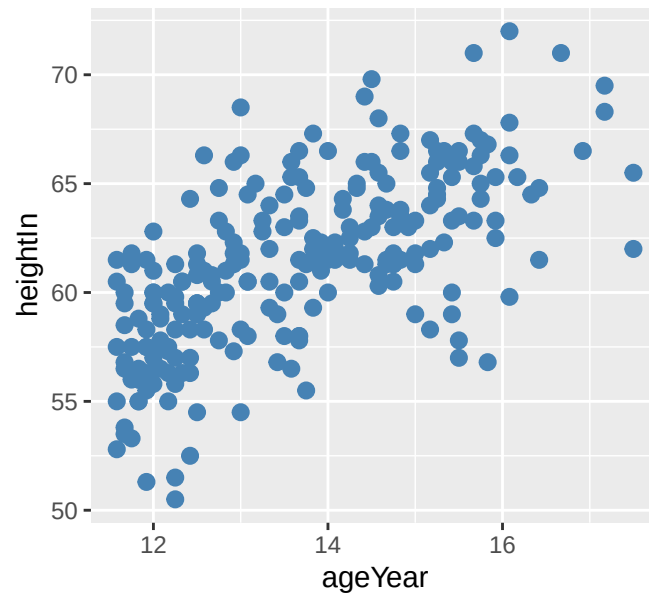


Figure 4.3: A scatterplot of Height by Year

We can add a regression line to the scatterplot:

```
p <- p + geom_smooth(method=lm, se=FALSE)
p
```

```
`geom_smooth()` using formula = 'y ~ x'
```

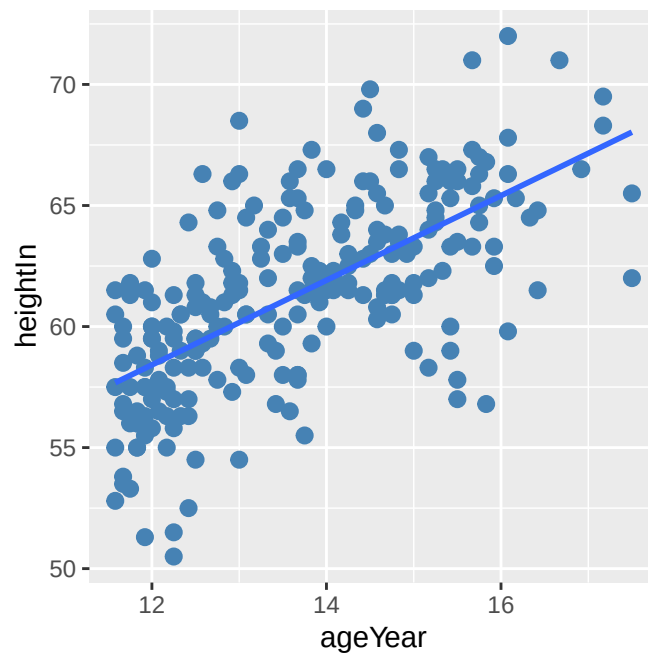


Figure 4.4: A scatterplot of Height by Year with a regression line

Fitting a simple linear regression Model

We fit a simple linear regression model to quantify the relationship between age and height:

```
mdl <- lm(heightIn ~ ageYear, data = heightweight)
summary(mdl)
```

Call:

```
lm(formula = heightIn ~ ageYear, data = heightweight)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.3517	-1.9006	0.1378	1.9071	8.3371

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	37.4356	1.8281	20.48	<2e-16 ***
ageYear	1.7483	0.1329	13.15	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.989 on 234 degrees of freedom

Multiple R-squared: 0.4249, Adjusted R-squared: 0.4225

F-statistic: 172.9 on 1 and 234 DF, p-value: < 2.2e-16

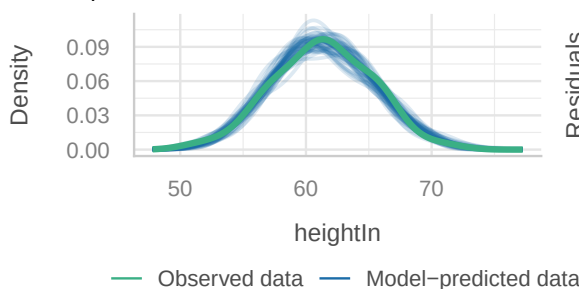
Assumptions

Linear regression has several assumptions (e.g., linearity, homoscedasticity, normality of residuals). We can check these assumptions using the `check_model()` function:

```
library(see)
library(patchwork)
library(performance)
check_model mdl
```

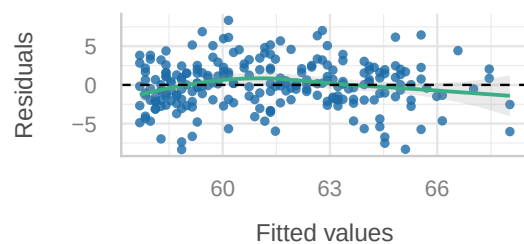
Posterior Predictive Check

Model-predicted lines should resemble observed data



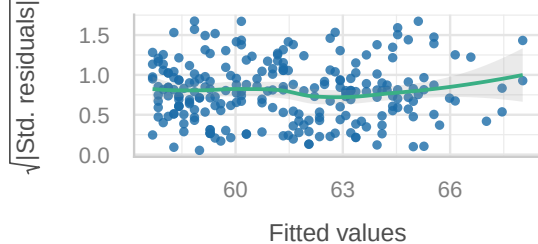
Linearity

Reference line should be flat and horizontal



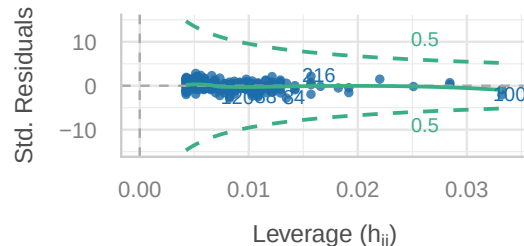
Homogeneity of Variance

Reference line should be flat and horizontal



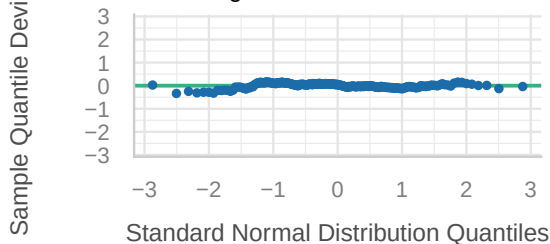
Influential Observations

Points should be inside the contour lines



Normality of Residuals

Points should fall along the line



4.8 Exercises

Exercise 8

Data: Verbeke & Van Praet (2025).

Is the average duration of the DRESS vowel different from 110 ms? Important: First, you need to take a subset from the data, so that you only look at the DRESS vowel. Use this code to create a subset:

```
dress <- subset(accent, Vowel == "DRESS") # "accent" is my object; you may have a different name
head(dress)
```

Exercise 9

The possessive alternation in English is illustrated by (1) and (2):

1. Kim's book
2. the book of Kim

The research question that we wish to answer is whether there is an association with the final sibilant. The following examples suggest that the s-genitive is less probable when the possessor ends in a sibilant sound:

3. the wheels of the bus
4. ?the bus's wheels

For this exercise, you examine `written_genitive_tokens_Brown_Frown.txt` from Grafmiller (2023). Open the data and select the variables `Type` and `Final_Sibilant`. Here is some example code that you could use:

```
gen <- read.delim("written_genitive_tokens_Brown_Frown.txt",
                 stringsAsFactors=TRUE)
gen <- gen[, c("Type", "Final_Sibilant")]
```

1. Crosstabulate `Type` and `Final_Sibilant`.
2. Evaluate the association between both variables by means of a chi-squared test.
3. Interpret your result.

Exercise 10

1. Open the dataset `Videolang.csv`.
2. Visualise the relation between `Posttest` and `Pretest` with a scatterplot.
3. fit a simple regression model with `Pretest` as a predictor and `Posttest` as the outcome.
4. Interpret the model estimates.

Chapter 5

NHST – Part 2

In this chapter, I provide a more in-depth theoretical explanation of the tests introduced in the previous chapter (although we're still scratching the surface of statistical inference theory). This is arguably the hardest chapter of the workshop and pretty challenging to grasp in a few hours time. If you prefer to keep things practical for the time being, then I suggest you skip this part. However, if you plan to perform more advanced statistical analyses and to understand what you're doing, some theoretical background is highly recommended.

5.1 Two-Sample t-Test

In the two-sample t-test, we compare the means of two independent samples to determine if there is a significant difference between them.

Comparing vocabulary scores

We create a mock dataset representing receptive vocabulary scores for the years 2000 and 2020. Our question is: Is there a significant difference between the mean scores in these two years?

We assume the following for our two populations:

- **2020:** $\mu = 66, \sigma = 16$
- **2000:** $\mu = 60, \sigma = 16$

```
set.seed(12)
Group <- as.factor(c(rep("2020", 60), rep("2000", 60)))
Score <- c(rnorm(n = 60, mean = 66, sd = 16), rnorm(n = 60, mean = 60, sd = 16))
Test <- data.frame(Group, Score)
head(Test) # the first six rows
```

	Group	Score
1	2020	42.31092
2	2020	91.23471
3	2020	50.69209
4	2020	51.27992
5	2020	34.03773
6	2020	61.64326

To explore the data, we first calculate basic descriptive statistics for each group:

```
library(tidyverse)

-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

Test |> group_by(Group) |>
  summarise(Mean = round(mean(Score), 2),
            VAR = round(var(Score), 2),
            SD = round(sd(Score), 2))

# A tibble: 2 x 4
  Group Mean  VAR  SD
  <fct> <dbl> <dbl> <dbl>
1 2000  61.5  197.  14.0
2 2020  64.4  197.  14.0
```

The distributions of the two groups can be visualized using a density plot:

```
ggplot(Test, aes(x = Score, fill = Group)) +
  geom_density(alpha = 0.6) +
  scale_fill_manual(values=c("#1E64C8", "#F1A42B")) +
  theme_minimal()
```

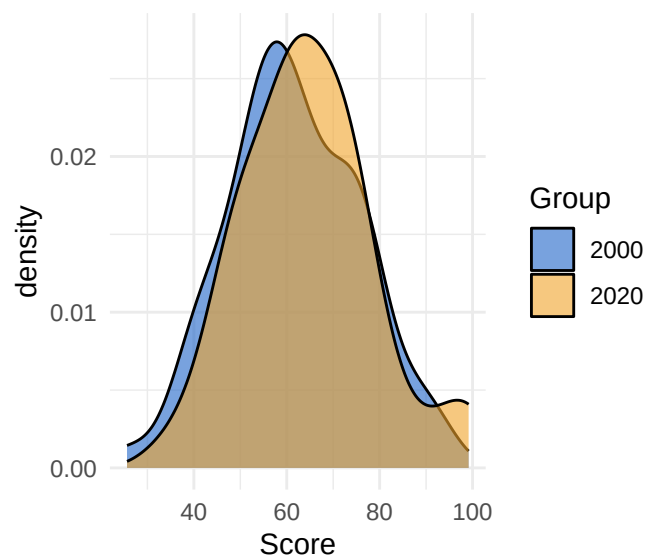


Figure 5.1: Score by Group

We now formulate the hypotheses that we will test by means of a t-test:

- H_0 : The means of the two groups are equal: $\mu_{2000} = \mu_{2020}$
- H_A : The means are different; $\mu_{2000} \neq \mu_{2020}$

Alternatively, the hypotheses could be formulated as directional:

- H_A : $\mu_{2000} > \mu_{2020}$
- H_A : $\mu_{2000} < \mu_{2020}$

Based on the test statistic and the associated P-value we make a decision: we either accept the null hypothesis or we reject the null hypothesis.

We reject the null hypothesis if:

- The test statistic is greater than the critical value.
- The p-value is smaller than the significance level (\$\$, typically 0.05).

The significance level α can be visualised as follows:

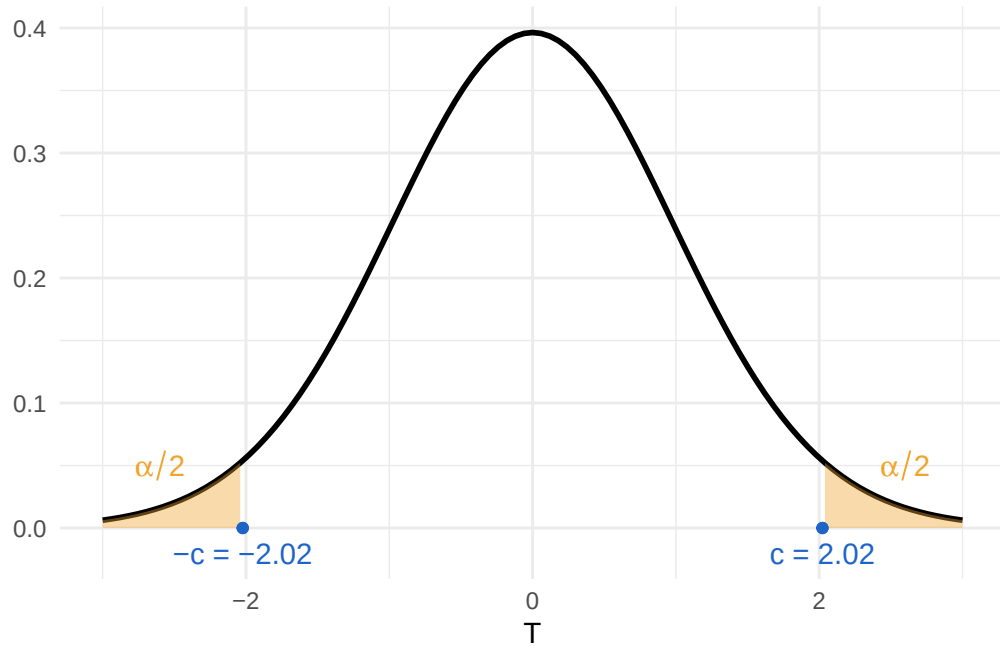


Figure 5.2: Significance level alpha

The two-sample t-test formula

The test statistic for the two-sample t-test is calculated as:

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2}$$

Where:

- $\bar{X}_1, \bar{X}_2$ are the sample means.

- s_p is the pooled standard deviation.
- n_1, n_2 are the sample sizes.

We perform the t-Test in R:

```
t.test(Score ~ Group, data = Test, alternative = "two.sided", var.equal = TRUE)
```

Two Sample t-test

data: Score by Group

t = -1.1357, df = 118, p-value = 0.2584

alternative hypothesis: true difference in means between group 2000 and group 2020 is not equal to 0

95 percent confidence interval:

-7.982992 2.163870

sample estimates:

mean in group 2000	mean in group 2020
61.52997	64.43953

The p-value helps us determine whether the observed difference between the two means is statistically significant:

```
pt(q = 1.13, df = 118, lower.tail = FALSE) * 2
```

```
[1] 0.260768
```

If the p-value is smaller than 0.05, we reject the null hypothesis.

The p-value can be visualised as follows:

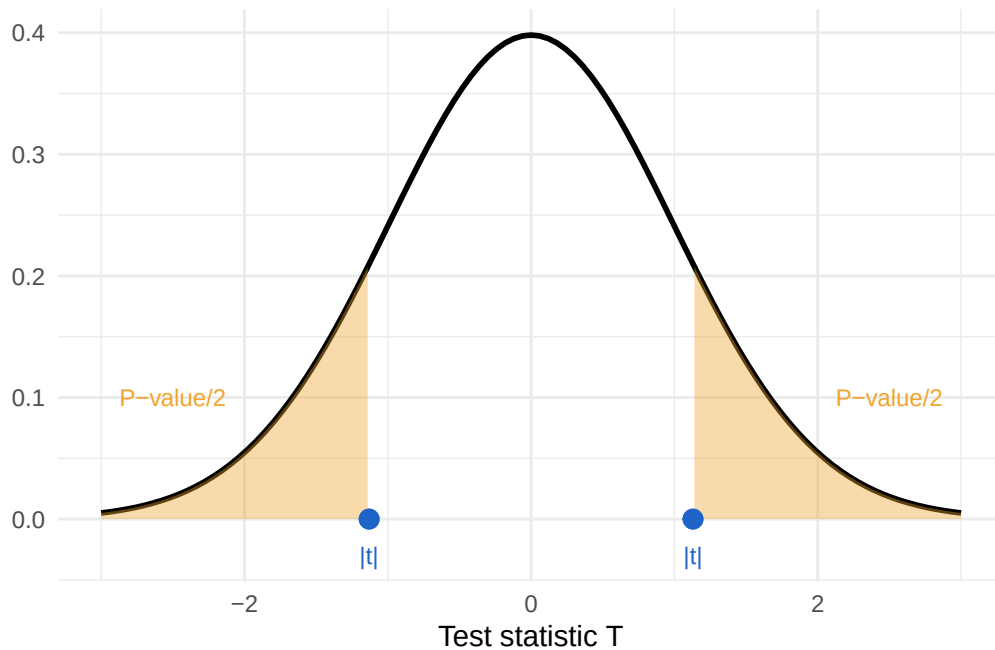


Figure 5.3: The P-value

The significance level α :

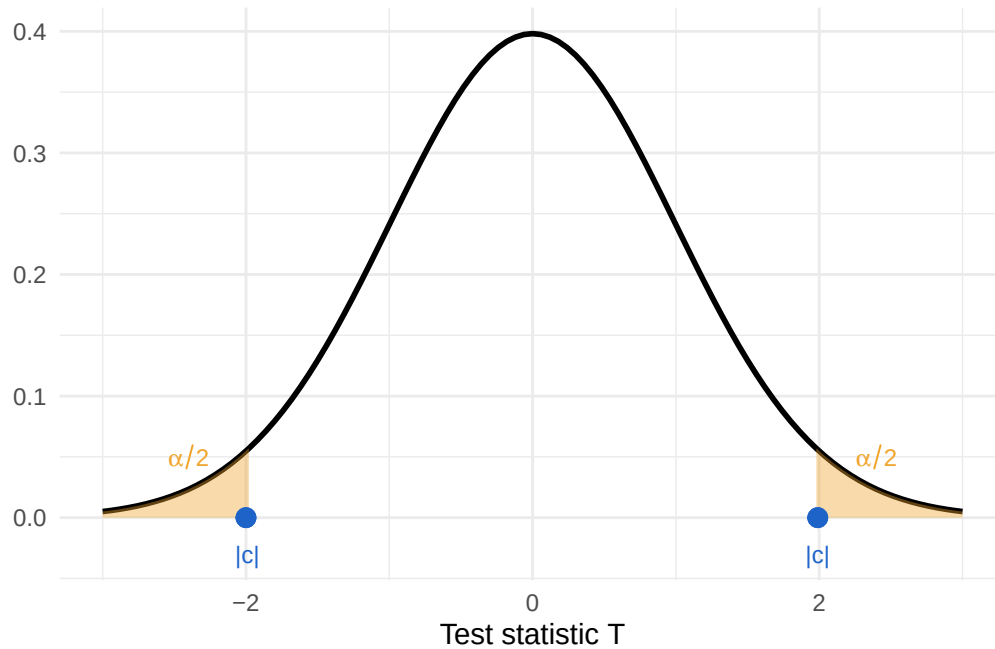


Figure 5.4: Significance level alpha

5.2 Chi-Squared Test

The chi-squared test is used to determine whether there is a significant association between two categorical variables.

Example: Possessive Alternation in English

We use data from Grafmiller ([grafmiller2023?](#)) to test whether there is a significant association between possessive construction type and the presence of a final sibilant. We load the data and select the two variables:

```
gen <- read.delim("written_genitive_tokens_Brown_Frown.txt", stringsAsFactors=TRUE)
gen <- gen[, c("Type", "Final_Sibilant")]
head(gen)
```

	Type	Final_Sibilant
1	s	N
2	of	N
3	of	N
4	s	N
5	of	Y
6	of	N

And crosstabulate the variables:

```
table("final sibilant" = gen$Final_Sibilant, "Type" = gen$Type)
```

```

      Type
final sibilant of  s
      N 2229 1754
      Y  874  241

```

We visualize the relationship between the two variables (here we use the tidyverse approach):

```

tt <- gen |>
  select(Final_Sibilant, Type) |>
  group_by(Final_Sibilant, Type) |>
  summarize(n = n())

```

`summarise()` has regrouped the output.
 i Summaries were computed grouped by Final_Sibilant and Type.
 i Output is grouped by Final_Sibilant.
 i Use `summarise(.groups = "drop\_last")` to silence this message.
 i Use `summarise(.by = c(Final\_Sibilant, Type))` for per-operation grouping
 (`?dplyr::dplyr\_by`) instead.

```

ggplot(data=tt, aes(x=Final_Sibilant, y=n, fill=Type)) +
  geom_col(position=position_dodge()) +
  scale_fill_manual('Type',
                    labels=c('of', 's'),
                    values=c('#F1A42B', '#1E64C8')) +
  scale_x_discrete(name="Final sibilant", labels=c("absent", "present")) +
  theme_minimal()

```

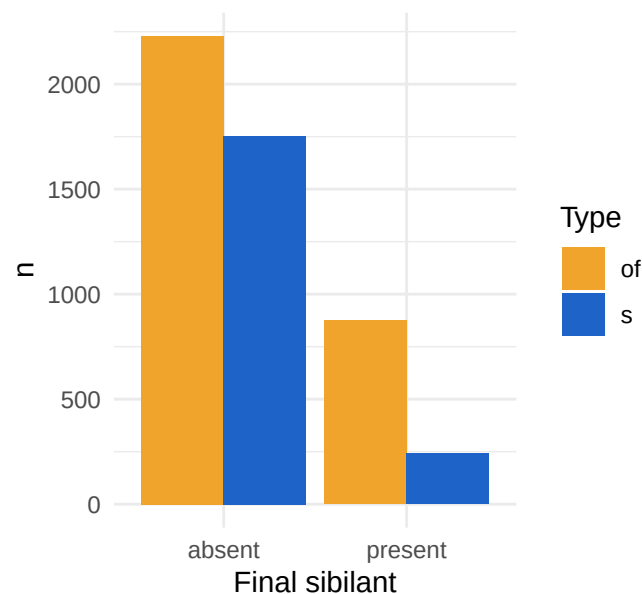


Figure 5.5: A clustered barplot of Type of Genitive by final sibilant

We formulate the null and alternative hypotheses:

- H_0 : the two variables are independent
- H_A : the two variables are not independent

The test statistic for the chi-squared test is calculated as:

$$X^2 = \sum \frac{(f_o - f_e)^2}{f_e}$$

Where f_o represents the observed frequencies and f_e represents the expected frequencies. We can calculate the expected frequencies using the following formula:

$$f_{e11} = \frac{\text{row 1 total} \times \text{column 1 total}}{\text{total number of observations}}$$

We extract the expected frequencies:

```
tt <- table(gen$Final_Sibilant, gen$Type)
chisq.test(tt)$exp
```

	of	s
N	2424.3329	1558.6671
Y	678.6671	436.3329

Let's calculate the test statistic by hand.

$$T = \frac{(2229 - 2424)^2}{2424} + \frac{(1754 - 1559)^2}{1559} + \frac{(874 - 679)^2}{679} + \frac{(241 - 436)^2}{436}$$

$$= 183$$

The test statistic is equal to 183. What is the probability of finding a test statistic that is even larger than this number (= the definition of the P-value)? Let's look at the visualisation of this probability (i.e., the p-value):

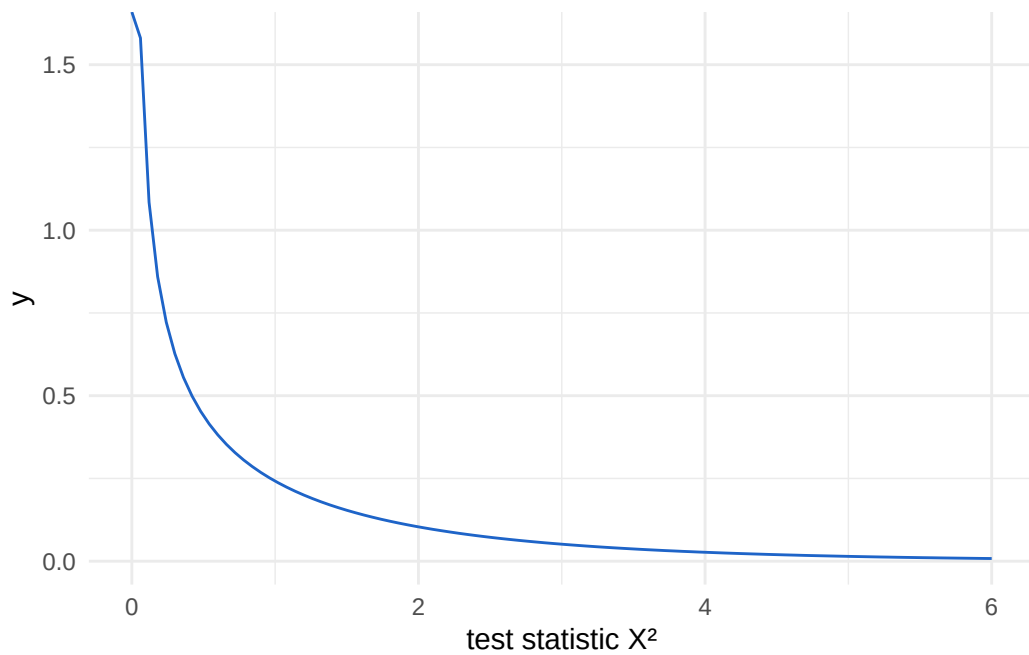


Figure 5.6: P-value

You need to look at the probability higher than the test statistic (183). This number is not shown on the plot, but you can see that the probability higher than 183 must be extremely low! The p-value is extremely low and thus highly significant. What is the critical value c associated with the significance level $\alpha = 5\%$?

```
qchisq(p = 0.05, df = 1, lower.tail = FALSE)
```

```
[1] 3.841459
```

183 is much larger!!

Let's look at the output of the chi-squared test in R again:

```
tt <- table(gen$Final_Sibilant, gen$Type)
chisq.test(tt)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: tt
```

```
X-squared = 182.94, df = 1, p-value < 2.2e-16
```

Figure Figure 5.7 shows more chi-squared distributions with different degrees of freedom:

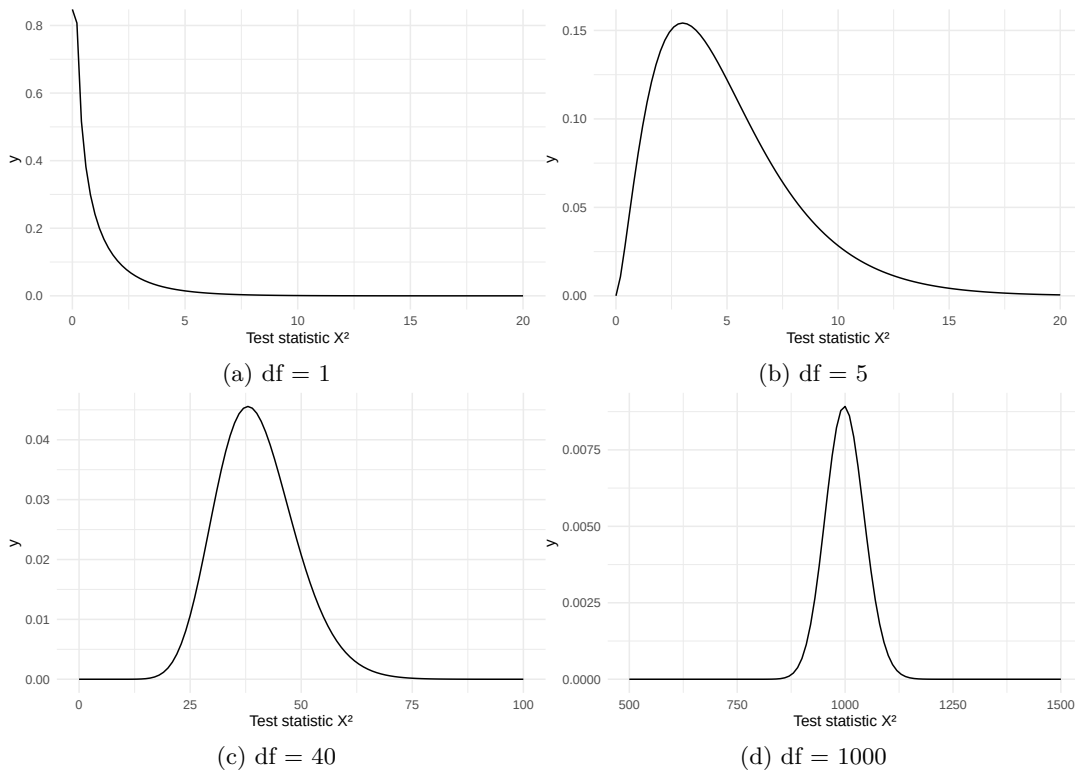


Figure 5.7: Chi-squared distributions

Effect Size: Cramér's V

It's important to note that strong evidence for an effect does not necessarily equal a strong effect. Cramér's V is a measure of effect size for chi-squared tests. It is calculated as:

$$V = \sqrt{\frac{X^2}{N \times \min(c - 1, r - 1)}}$$

Where:

- N = number of observations
- c = number of columns
- r = number of rows
- $\min(c - 1, r - 1)$: the smallest number of the couple

The outcome of Cramér's V is a percentage. The following rules of thumb apply:

- $V = 1$: Perfect association.
- $V = 0$: No association.
- $V < 0.25$: Weak association.
- $0.25 < V < 0.75$: Moderate association.
- $V > 0.75$: Strong association.

We calculate Cramér's V for our example at hand:

$$\begin{aligned}
 V &= \sqrt{\frac{X^2}{N * \min(c - 1, r - 1)}} \\
 &= \sqrt{\frac{183}{5098 * \min(2 - 1, 2 - 1)}} \\
 &= \sqrt{\frac{183}{5098}} \\
 &= 0.19
 \end{aligned}$$

The effect size is rather low, which suggests that the final sibilant only explains a small part of the variability associated with the possessive alternation (i.e., other variables are likely to be involved as well).

5.3 Exercise

For this exercise, you will use the **FalseBeginners** dataset from De Wilde (2019). The dataset contains information on language proficiency tests and student attitudes toward English.

ID	Variable	Description
1	Student	A unique identifier for each student
2	School	A unique identifier for each school
3	Class	A unique identifier for each class
4	PPVT	Score for the Peabody Picture Vocabulary Test (PPVT). Maximum score = 120
5	Speaking	Score for the speaking test. Maximum score = 20
6	Listening	Score for the listening test. Maximum score = 25
7	ReadingWriting	Score for the reading and writing tests. Maximum score = 50
8	Attitude	Attitude of student towards English: Positive vs. Negative
9	L1	L1: Dutch vs. Multilingual
10	Sex	Sex of student: Male vs. Female

1. Load the **Falsebeginners** dataset.
2. Provide a univariate summary of every variable.
3. Visualize the distribution of PPVT scores using a blue histogram.
4. Visualize the distribution of PPVT scores using a green boxplot.
5. Create a clustered boxplot of PPVT scores by Sex.
6. Visualize the distribution of PPVT scores by Sex using overlapping histograms in ggplot2.
7. Perform a t-test to determine if there is a significant difference in PPVT scores between male and female students.
8. Create a table summarizing the relationship between L1 and Sex.
9. Use a chi-squared test to assess whether L1 is associated with Sex.
10. Create a scatterplot to visualize the relationship between Speaking and Listening scores, and add a regression line.

11. Fit a simple linear regression model with Speaking as the predictor and Listening as the outcome variable.

References

- De Wilde, V., Brysbaert, M., & Eyckmans, J. (2019). Learning english through out-of-school exposure. Which levels of language proficiency are attained and which types of input are important? *Bilingualism: Language and Cognition*, 23(1), 171–185. <https://doi.org/10.1017/s1366728918001062>
- Grafmiller, J. (2023). *The genitive alternation in 1960s and 1990s American English: Data from the Brown and Frown corpora* (Version V1) [dataset]. DataverseNO. <https://doi.org/10.18710/R7HM8J>
- Verbeke, G., & Van Praet, W. (2025). *Replication Data for: The interplay between prosodic prominence and boundary strength in the production of English checked steady-state vowels* (Version V2) [dataset]. DataverseNO. <https://doi.org/10.18710/QRXULF>

